CWL toolkit for single-cell sequencing data analysis

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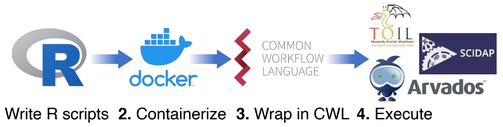
https://github.com/Barski-lab/sc-seq-analysis
https://github.com/datirium/workflows

Abstract

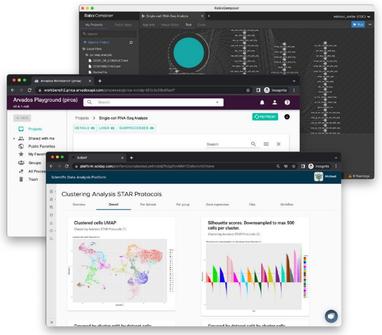
The massive growth in the amount of research data, and in particular data generated in single-cell experiments, created a huge demand on highly efficient bioinformatics tools and workflows. It also underscored two important problems in data science – **poor reproducibility and limited portability of data analysis methods and techniques**. These issues are very important to address because only accurate and reproducible results of bioinformatics data analyses may lead to new achievements in science. Reproducibility becomes of particular importance when it comes to single-cell sequencing data analysis where even the slightest change in the input parameters or software versions may produce previously unseen results which require extra time and efforts for validation. To overcome the reproducibility and portability challenges **we develop all our pipelines using Common Workflow Language** (CWL) as it is designed for data-intensive science and helps us to follow FAIR (Findable, Accessible, Interoperable and Reusable) paradigms.



1. Write R scripts **2.** Containerize **3.** Wrap in CWL **4.** Execute

Advantages of using CWL

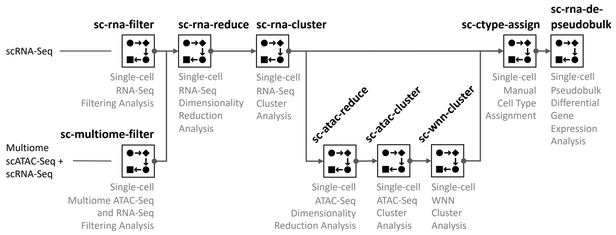
CWL is an open standard that describes tools and pipelines as YAML or JSON structured linked data documents. For each workflow step ("tool"), CWL defines inputs and outputs in a formalized way that allows for combining of tools into workflows. Moreover, CWL tool description is linked to Docker (or Singularity) container supporting execution in an isolated runtime environment with a pre-installed and tested version of the required software. Therefore, **our pipelines and tools are independent of the execution platform** and can be run in exactly the same way by any workflow runner that supports CWL standard.



- ✓ **SciDAP platform**
  - CWL-Airflow + Docker
- ✓ **Arvados Playground**
  - arvados-cwl-runner + Docker
- ✓ **Rabix Composer**
  - Rabix Executor + Docker
- ✓ **HPC cluster**
  - toil-cwl-runner + Singularity
- ✓ **cwltool**
  - Docker
  - Singularity

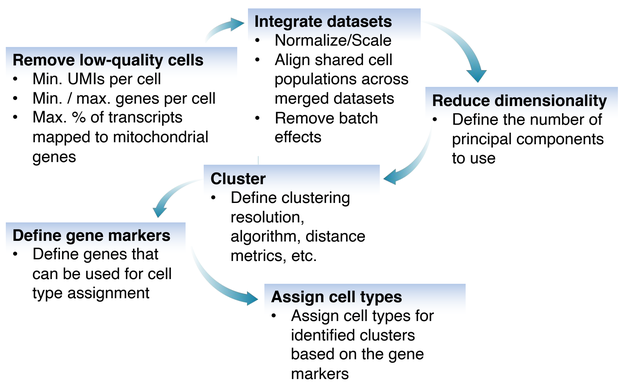
Build your own workflow

Here we present a set of CWL tools and pipelines for preprocessing and analysis of single-cell RNA, ATAC and Multiome sequencing data. Altogether they allow to execute the most common tasks in this research area such as low-quality cells removal, data integration, batch effect correction, clustering, cell type assignment, and differential gene expression analysis. Each tool produces an extensive set of publication ready PDF/PNG figures and CSV/TSV tables for results assessment and interpretation. Furthermore, the outputs are also saved in R compatible RDS format which allows to continue analysis if needed. These **tools can be chained together to form various workflows for a particular research application.**



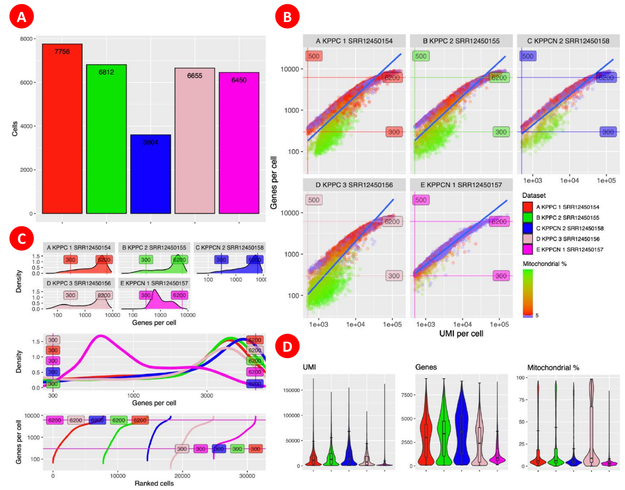
Overview of scRNA-Seq analysis

In most of the cases scRNA-Seq data analysis is an iterative process that requires multiple reruns to adjust filtering, integration, and clustering parameters. Being able to execute the whole pipeline or its individual step multiple times and obtain the same results is particularly important when measuring the influence of each specific parameter. To achieve such level of reproducibility all our **R scripts are encapsulated in Docker containers**. Additionally, most of our CWL tools produce required QC metrics to assess the analyses results and potentially avoid unneeded reruns.

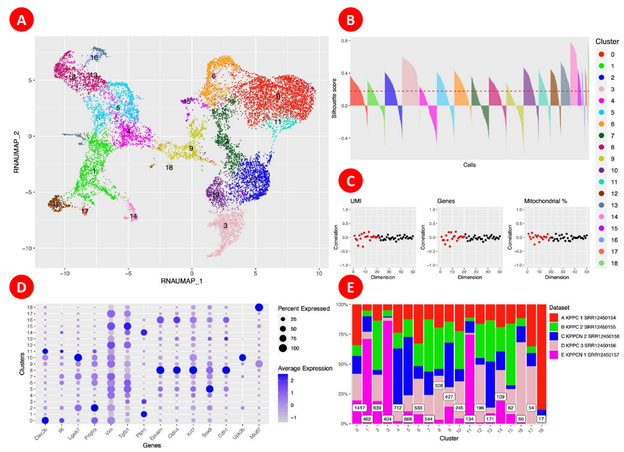


scRNA-Seq application

As an example of a successful workflow application, we analyzed scRNA-Seq data from the study related to the differentiation of pancreatic ductal adenocarcinoma (PDAC) *(Surumbayeva, Kotliar et al., 2021*). In the original paper (*Gabitova-Cornell et al., 2020*), data analysis was conducted by manual command line and R processing thus limiting reproducibility and portability of computational approaches. Having all of the pipelines in the CWL format allows us to **run them in any CWL-based execution environment**. However, we prefer using specialized data analysis platform SciDAP as it helps us to keep data organized and provides a convenient graphical user interface.



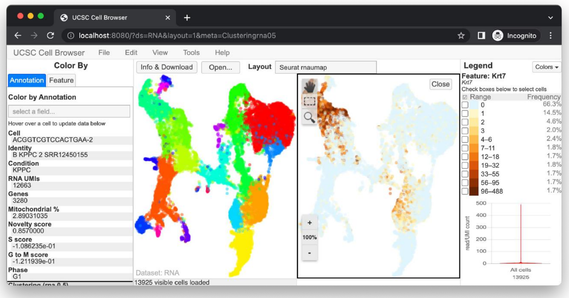
Cell counts (A), genes vs UMIs per cell correlation (B), genes per cell density distribution (C), and QC metrics violin plots (D) per dataset.



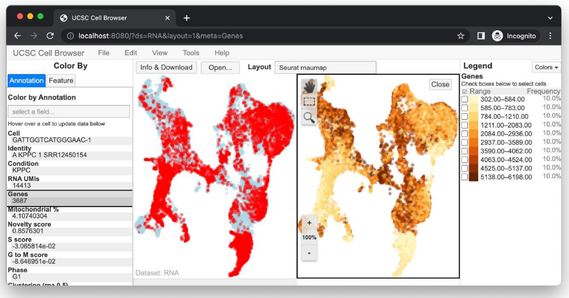
Clustering (A), silhouette scores (B), PC vs QC metrics correlation (C), average gene expression per cluster (D), and composition (E) plots.

Interactive visualization

Clustering results loaded in UCSC Cell Browser allow to interactively explore identified clusters, highlight cells belonging to various samples and conditions, explore expression of genes of interest, and export barcodes of selected cells for further analyses.



UCSC Cell Browser: clustering with resolution 0.5 (left), Krt7 gene expression (right)



UCSC Cell Browser: grouping by biological condition (left), genes per cell QC metrics (right)

References

- Surumbayeva, A., Kotliar, M., Gabitova-Cornell, L., Kartashov, A., Peri, S., Salomonis, N., Barski, A., Astsaturov, I. (2021). Preparation of mouse pancreatic tumor for single-cell RNA sequencing and analysis of the data. *STAR Protocols*, 2(4), 100989, ISSN 2666-1667.
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